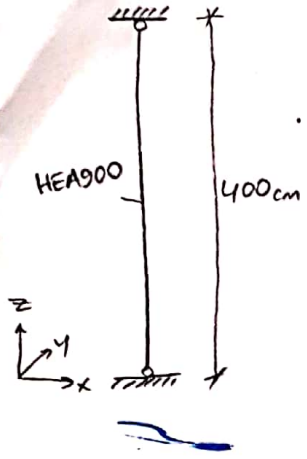


A compression member formed by HEA900 is pinned at both end for X-Z and Y-Z axes. Calculate design compression strength of the member.

Steel: S355

$F_y = 35,5$ kN/cm²

$F_u = 51$ kN/cm²

$E = 20000$ kN/cm²

HEA900

$d = 890$ $\quad I_x = 422100$ cm⁴

$b = 300$ $\quad i_x = 36,29$ cm

$t_w = 16$ $\quad I_y = 13550$ cm⁴

$t_f = 30$ $\quad i_y = 6,50$ cm

$A = 320,5$ cm²

$h = 770$

• Local buckling control

→ flanges: $\lambda = \frac{b}{2t_f} = \frac{300}{2 \cdot 30} = 5 < \lambda_r = 0,56\sqrt{\frac{E}{F_y}} = 13,29$ non-slender

→ web: $\lambda = \frac{h}{t_w} = \frac{770}{16} = 48,13 > \lambda_r = 1,49\sqrt{\frac{E}{F_y}} = 35,37$ slender

$\lambda \le \lambda_r \sqrt{\frac{F_y}{F_{cr}}} \rightarrow b_e = b$

$\lambda > \lambda_r \sqrt{\frac{F_y}{F_{cr}}} \rightarrow b_e = b\left(1 - c_1\sqrt{\frac{F_{el}}{F_{cr}}}\right)\sqrt{\frac{F_{el}}{F_{cr}}}$

λ = slenderness ratio of slender element (flange or web)

b = width of slender element (flange or web)

b_e = effective with (d_e for T, h_e for web)

$\lambda_x = \frac{L_{cx}}{i_x} = \frac{400}{36,29} = 11,02$

$\lambda_y = \frac{L_{cy}}{i_y} = \frac{400}{6,50} = 61,54$

$\lambda_{max} = 61,54 < 4,71\sqrt{\frac{E}{F_y}} = 111,79$ inelastic buckling

$F_e = \frac{\pi^2 E}{\lambda_{max}^2} = \frac{\pi^2 \cdot 20000}{61,54^2} = 52,12$ kN/cm²

$F_{cr} = 0,658^{F_y/F_e} \cdot F_y = 0,658^{35,5/52,12} \cdot 35,5 = 26,69$ kN/cm²

$\lambda = 48,13 > 35,37\sqrt{\frac{35,5}{26,69}} = 40,79$

$F_{el} = \left(c_2 \frac{\lambda_r}{\lambda}\right)^2 F_y$: elastic local buckling stress

c_1, c_2 : taken table 8.2

$c_1 = 0,18$

$c_2 = 1,31$

$F_{el} = \left(1,31 \cdot \frac{35,37}{48,13}\right)^2 \cdot 35,5 = 32,9$ kN/cm²

$h_e = 770\left(1 - 0,18\sqrt{\frac{32,9}{26,69}}\right)\sqrt{\frac{32,9}{26,69}} = 684,05$

CamScanner ile tarandı

$A_e = A_g - (h - h_e).t_w = 320,5 - (77 - 68,41).1,6 = 306,76\ cm^2$

$P_n = F_{cr}.A_e = 26,69.306,76 = 8187,42\ kN \qquad \phi P_n = 7368,68\ kN$

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